

PROGRAM - 6

TITLE: Deploy an Automated build pipeline using Docker Hub

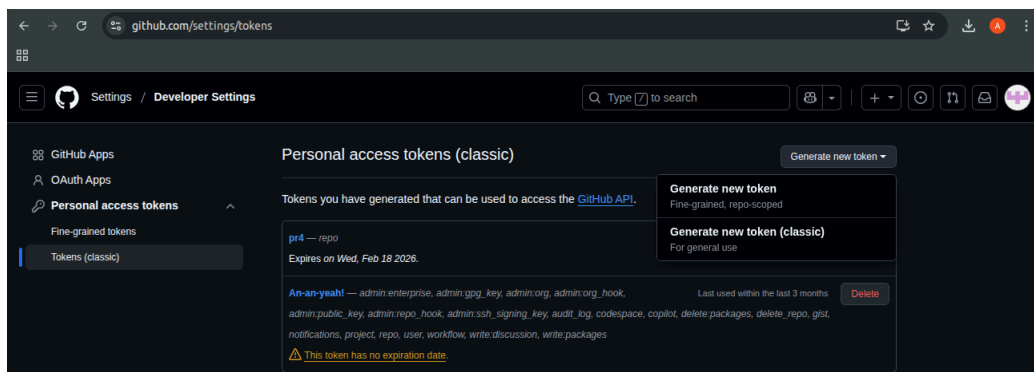
Project Structure

Program-6

- Dockerfile
- node_modules
- package.json
- Package-lock.json

Step 1: Create a PAT and copy the password

Settings—> Developer Settings—> Personal Access Token—> Classic Tokens—> Create token



Step-2: Create a Dockerfile

```
GNU nano 6.2
FROM node:18-alpine
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node","start"]
```

Step-3: create package.json by executing the command

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ npm init -y
Wrote to /home/user/labprograms/Program-6/package.json:
```

```
{
  "name": "program-6",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step-4: create package-lock.json by executing the command

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ npm install

up to date, audited 1 package in 296ms

found 0 vulnerabilities
```

Step-4: Build docker image

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker build -t prg6 .
[+] Building 35.0s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 150B                                0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  2.3s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd 26.0s
=> => resolve docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3 0.0s
=> => sha256:f18232174bc91741fdf3da96d85011092101a032a93 3.64MB / 3.64MB 2.4s
=> => sha256:dd71dde834b5c203d162902e6b8994cb2309ae04 40.01MB / 40.01MB 24.5s
=> => sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda0183 1.26MB / 1.26MB 4.9s
=> => sha256:8d6421d663b4c28fd3ebc498332f249011d11894558 7.67kB / 7.67kB 0.0s
=> => sha256:929b04d7c782f04f615cf785488fed452b6569f87c7 1.72kB / 1.72kB 0.0s
=> => sha256:ee77c6cd7c1886ecc802ad6cedef3a8ec1ea27d1fb9 6.18kB / 6.18kB 0.0s
=> => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b 0.1s
=> => sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9 446B / 446B 4.2s
=> => extracting sha256:dd71dde834b5c203d162902e6b8994cb2309ae049a0eabc4 1.2s
=> => extracting sha256:1e5a4c89cee5c0826c540ab06d4b6b491c96eda01837f430 0.1s
=> => extracting sha256:25ff2da83641908f65c3a74d80409d6b1b62ccfaab220b9e 0.0s
=> [internal] load build context                                   0.0s
=> => transferring context: 669B                                       0.0s
=> [2/5] WORKDIR /app                                              0.4s
=> [3/5] COPY package*.json ./                                     0.1s
=> [4/5] RUN npm install                                           5.9s
=> [5/5] COPY . .                                                  0.1s
=> exporting to image                                              0.1s
=> => exporting layers                                              0.1s
=> => writing image sha256:d93737d15bcce424f8cfd665bbe29385cf027950a7bf0 0.0s
```

Step-5: tag the image for docker hub

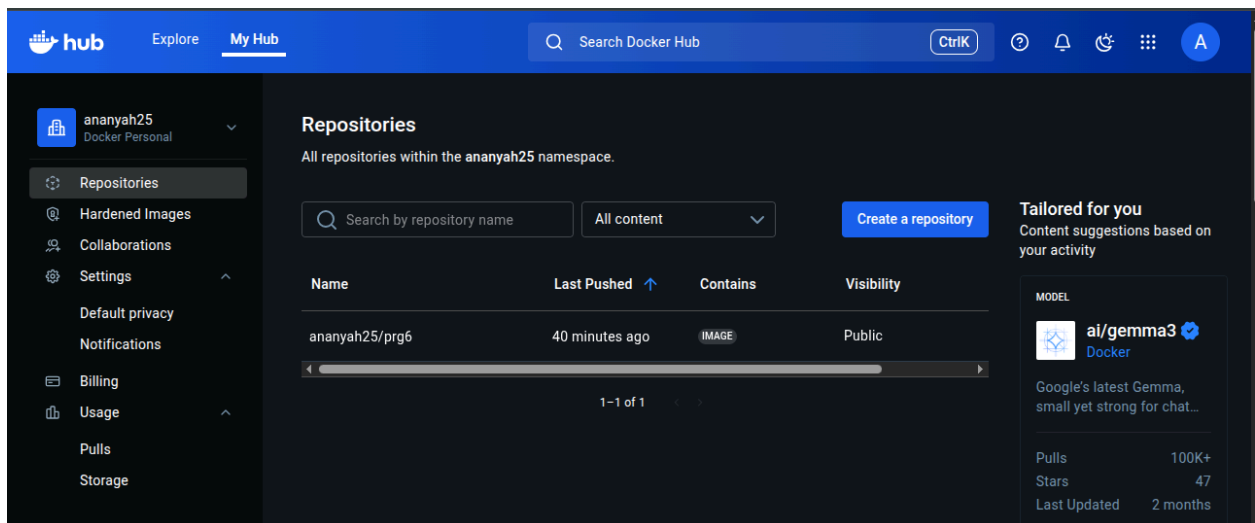
```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker tag prg6 ananyah25/prg6:latest
```

Step-6: login and push the image to docker hub

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker login -u ananyah25

Info → A Personal Access Token (PAT) can be used instead.
       To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker push ananyah25/prg6:latest
The push refers to repository [docker.io/ananyah25/prg6]
5cdd39bdc8ee: Pushed
53662de372de: Pushed
b28d4f7f6a40: Pushed
03d6d298e4fc: Pushed
82140d9a70a7: Mounted from library/node
f3b40b0cdb1c: Mounted from library/node
0b1f26057bd0: Mounted from library/node
08000c18d16d: Mounted from library/node
latest: digest: sha256:e061d8381c8f8d904059e66907ba237d4e15ecf39b651f8b9f2fa1242a070334 size: 1985
```



Step-7: Tag the image for Github container repository

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker tag prg6 ghcr.io/ananyah25/prg6:latest
```

Step-8: login and Push the image to github container registry

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker tag prg6 ghcr.io/ananyah25/prg6:latest
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker login ghcr.io -u ananyah25
Password:
Login Succeeded
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ docker push ghcr.io/ananyah25/prg6:latest
The push refers to repository [ghcr.io/ananyah25/prg6]
5cdd39bdc8ee: Pushed
53662de372de: Pushed
b28d4f7f6a40: Pushed
03d6d298e4fc: Pushed
82140d9a70a7: Pushed
f3b40b0cdb1c: Pushed
0b1f26057bd0: Pushed
08000c18d16d: Pushed
latest: digest: sha256:e061d8381c8f8d904059e66907ba237d4e15ecf39b651f8b9f2fa1242a070334 size: 1985
```

Step-9: Initialize the git repository and push the source code

- > git init
- > git add .
- > git commit -m "program-6"
- > git remote add https://github.com/Ananyah25/Program-4.git
- > git branch -m main
- > git push -u origin main

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/user/labprograms/Program-6/.git/
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git add .
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git commit -m "program-6"
[master (root-commit) 51c384f] program-6
 3 files changed, 32 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 package-lock.json
 create mode 100644 package.json
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git remote add origin https://github.com/Ananyah25/Program-6.git
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git branch -M main
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$ git push -u origin main
Username for 'https://github.com': Ananyah25
Password for 'https://Ananyah25@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 677 bytes | 677.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ananyah25/Program-6.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-6$
```

Ananyah25

Q Type [7] to search

+

Overview

Repositories 12

Projects

Packages

Stars

Ananyah25

Edit profile

Type: All

Q Search packages...

Visibility: All

Sort by: Most downloads

1 package

prg6

Private

Published 27 minutes ago by Ananyah25

0

Ananyah25 / Program-6

Q Type [7] to search

+

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Program-6

Public

Pin

Watch 0

Fork 0

Star 0

main

1 Branch

0 Tags

Go to file

Add file

Code

Ananyah25

program-6

51c384f · 28 minutes ago

1 Commit

Dockerfile

program-6

28 minutes ago

package-lock.json

program-6

28 minutes ago

package.json

program-6

28 minutes ago

README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

